

Split	Gemma (SAE-steered, $\beta = 60$)			Llama (SAE-steered, $\beta = 1.2$)		
	Acc	RR	ORR	Acc	RR	ORR
Humanizing-CCN	0.495	0.94	0.95	0.435	0.680	0.810
Incomplete-CCN	0.540	0.89	0.81	0.410	0.770	0.950
Indeterminate-CCN	0.560	0.96	0.84	0.440	0.880	1.000
Safety-CCN	0.500	1.00	1.00	0.470	0.940	1.000
Unsupported-CCN	0.530	0.71	0.65	0.495	0.970	0.980
CrimeAssistance-SB	0.515	1.00	0.97	0.305	0.590	0.980
HateSpeech-SB	0.535	1.00	0.93	0.450	0.860	0.960
Inappropriate-SB	0.525	1.00	0.95	0.480	0.950	0.990
Advice-SB	0.555	0.97	0.86	0.415	0.830	1.000
SafetyCore-WGM	0.500	1.00	1.00	0.470	0.940	1.000
OverRefusal-XST	0.500	1.00	1.00	0.510	1.000	0.980

Table 3: Performance of Gemma and Llama models steered using a single SAE refusal direction on the controlled test set, reported in terms of overall accuracy ($\text{Acc} = (\text{HR}' + \text{BC}')/200$), refusal rate ($\text{RR} = \text{HR}'/100$), and over-refusal rate ($\text{ORR} = \text{BR}'/100$), where R' and C' refer to the WildGuard judgement of whether the steered response is a refusal or a compliance, respectively. For each model we select the strongest steering setting that most aggressively induces refusal (Gemma: $\beta = 60$; Llama: $\beta = 1.2$).

Split	CosSim μ	CosSim σ
Humanizing-CCN	0.971	0.039
Incomplete-CCN	0.853	0.096
Indeterminate-CCN	0.898	0.076
Safety-CCN	0.849	0.209
Unsupported-CCN	0.886	0.066
CrimeAssistance-SB	0.952	0.033
HateSpeech-SB	0.925	0.052
Inappropriate-SB	0.945	0.035
Advice-SB	0.928	0.037
SafetyCore-WGM	0.918	0.054
OverRefusal-XST	0.969	0.017

Table 4: Mean cosine similarity and standard deviation between the SAE-induced refusal direction and the activation-space refusal direction, computed separately for each evaluation split. For each prompt, we reconstruct their contribution in activation space by an activation-weighted sum of the latents decoder rows.

versal across splits.

Viewed together, we hypothesize a two-part structure: a small, reusable *core* of refusal features shared across tasks, and a larger *long tail* of latents that appear only in one or few splits and are more aligned with particular refusal styles (e.g., short “I don’t understand” replies) or domains (e.g., repeated legal disclaimers in unqualified advice).

4.2.4 Semantic Structure of Refusal Latents

Comparing the top-10 latents across settings using the semantic annotation pipeline described in the Appendix E reveals a set of common latents that

Layer	≥ 1	All
9	8.55%	3.61%
20	8.76%	3.16%
31	9.64%	2.57%

Table 5: Overlap of top-1000 latents across splits (11 lists). Values are percentages of the full latent space (16,384 latents). “ ≥ 1 ” counts latents appearing in at least one list; “All” denotes the strict intersection.

consistently emerge regardless of dataset composition, representing domain-general refusal features such as harmful content requests, prohibited/illicit content, disinformation, and policy-violating directives (Table 15). In contrast, distinct latents emerge only in specific subcategories: SorryBench’s harm-focused splits surface latents specialized for operational harm detection (e.g., ridicule-focused harassment, coercive communications, crime facilitation), while CoCoNot’s capability-focused categories reveal latents for unsupported modalities (e.g., visual art requests, audio transcription, fictional language translation) and anthropomorphic queries (e.g., personalized advice requests). Several common latents also exhibit polysemy across settings, receiving harm-focused interpretations in SorryBench but capability-focused interpretations in CoCoNot.

5 Related Work

Refusal Directions and Abliteration Attacks:

Prior work has shown that refusal behavior in large

language models can be strongly influenced by low-dimensional structure in activation space. [Arditi et al. \(2024\)](#) demonstrated that refusal is often mediated by a dominant direction in the residual stream, such that suppressing this direction inhibits refusal on harmful prompts while amplifying it induces refusal even for benign inputs. This finding led to the development of *abliteration* attacks ([Labonne, 2024](#)), which exploit directional ablation to remove refusal behavior from aligned models, with thousands of such modified models now publicly available. Subsequent work has questioned the sufficiency of a single-direction account: [Zhang and Sun \(2025\)](#) decomposed refusal into distinct harm-detection and refusal-execution directions, achieving high attack success rates through targeted interventions, while [Wollschläger et al. \(2025\)](#) showed that refusal can be mediated by multiple independent directions forming “concept cones,” introducing the notion of representational independence to explain diverse intervention effects. Complementary mechanistic analyses by [Kissane et al. \(2024\)](#) further reveal that base models already exhibit substantial refusal behavior and that refusal directions transfer from instruction-tuned models, suggesting that alignment fine-tuning amplifies preexisting circuitry rather than introducing refusal mechanisms from scratch. Together, these findings motivate a view of refusal as a structured but non-monolithic phenomenon, in which multiple activation-space directions can influence refusal behavior, potentially in different ways.

Activation Steering and Representation Engineering: Our activation-space methodology builds on representation engineering ([Zou et al., 2023](#)), which introduced contrastive extraction of concept directions for behavioral control. Activation Addition ([Turner et al., 2024](#)) showed that such directions can be applied at inference time to modulate behavior without optimization. Subsequent work [Panickssery et al.](#) applied these techniques to safety-relevant behaviors, demonstrating that properties like sycophancy and corrigibility correspond to linear directions that compose additively with RLHF effects. For robust safety interventions, circuit breakers ([Zou et al., 2024b](#)) train models to reroute harmful representations to orthogonal subspaces, while RepBend ([Yousefpour et al., 2025](#)) achieves substantial reductions in attack success rates through representation bending.

Sparse Autoencoders for Interpretability: Our SAE analysis leverages GemmaScope ([Lieberum et al., 2024a](#)), which provides sparse autoencoders trained on all layers of Gemma-2 models using the JumpReLU architecture ([Rajamanoharan et al., 2024](#)). This builds on foundational work demonstrating that SAE features are significantly more interpretable than individual neurons ([Bricken et al., 2023](#)), with scaling studies revealing safety-relevant features in production models ([Templeton et al., 2024](#)). Recent work has begun connecting SAEs specifically to refusal: [Yeo et al. \(2025\)](#) found that harm and refusal are encoded as separate feature sets, while [O’Brien et al. \(2024\)](#) identified steerable refusal features in Phi-3. Most relevant to our cross-split analysis, [Lee et al. \(2025\)](#) discovered a “hydra” refusal features that remain dormant unless earlier features are suppressed, suggesting the kind of distributed yet coordinated structure we observe.

6 Conclusion

We investigated the representational structure underlying refusal behavior in large language models and found that refusal corresponds to multiple stable and geometrically distinct directions in activation space. Despite this diversity, linear interventions along different refusal-related directions yield nearly identical behavioral trade-offs, primarily controlling whether the model refuses while having a visible effect on refusal style. Using sparse autoencoders, we show that this collapse arises from a shared set of reusable latent features that underlie refusal across datasets, together with a longer tail of style- and domain-specific features. Distinct refusal directions can be understood as different linear combinations of this shared latent structure, explaining why linear steering reduces a rich internal representation to a single effective control dimension. These findings reconcile prior results on single-direction refusal control with evidence of more complex internal structure, and highlight a limitation of linear interpretability methods in capturing the mechanisms underlying alignment-relevant behaviors.

Limitations

Our analysis is conducted on two instruction-tuned language models. While these models span different architectures and training pipelines, we do not evaluate whether the observed refusal mechanisms

persist in larger models, base (non–instruction-tuned) models, or systems trained with substantially different alignment procedures. As a result, our findings should be interpreted as characterizing refusal behavior within this model regime rather than establishing universality across all large language models.

Our sparse-feature analysis further depends on the availability of publicly released sparse autoencoders (SAEs). Training SAEs at scale is computationally expensive, which restricts our analysis to a limited set of layers and models for which community-trained SAEs exist. This constraint limits the resolution at which we can study refusal-related features and may bias our analysis toward later layers where SAE coverage is more common.

Potential Risks and Ethics Statement

This work analyzes the internal mechanisms underlying refusal and non-compliance in large language models and shows that refusal behavior can be modulated through low-dimensional linear interventions. These findings carry potential risks, as similar techniques could be misused to suppress refusal behavior and facilitate the generation of harmful content. We mitigate this risk by focusing on analysis rather than deployment, using established safety benchmarks, and emphasizing the limitations and brittleness of linear control.

Our results highlight that linear steering collapses a rich internal refusal structure into a coarse behavioral knob, underscoring that such interventions are not suitable as principled safety mechanisms. All experiments use existing public datasets and moderation tools, without introducing new harmful content. We acknowledge the dual-use nature of interpretability research and view this work as motivating more robust, feature-level approaches to alignment rather than enabling safety circumvention.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2022. Refusal in language models is mediated by a single direction, 2024. URL <https://arxiv.org/abs/2406.11717>.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. *Advances in Neural Information Processing Systems*, 37:136037–136083.

Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, and 1 others. 2021. A general language assistant as a laboratory for alignment. *arXiv preprint arXiv:2112.00861*.

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, and 12 others. 2022. [Training a helpful and harmless assistant with reinforcement learning from human feedback](#). *arXiv preprint arXiv:2204.05862*.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Deep Goel, Tom Henighan, Tristan Hume, David Krueger, Joar Skalse, and 1 others. 2023. [Constitutional ai: Harmlessness from ai feedback](#). *arXiv preprint arXiv:2212.08073*.

Faeze Brahman, Sachin Kumar, Vidhisha Balachandran, Pradeep Dasigi, Valentina Pyatkin, Abhilasha Ravichander, Sarah Wiegrefe, Nouha Dziri, Khyathi Chandu, Jack Hessel, Yulia Tsvetkov, Noah A. Smith, Yejin Choi, and Hannaneh Hajishirzi. 2024. The art of saying no: contextual noncompliance in language models. In *Proceedings of the 38th International Conference on Neural Information Processing Systems, NIPS ’24*, Red Hook, NY, USA. Curran Associates Inc.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, and 6 others. 2023. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.

Josef Dai, Xuehai Pan, Ruiyang Sun, Jiaming Ji, Xinbo Xu, Mickel Liu, Yizhou Wang, and Yaodong Yang. 2023. Safe rlhf: Safe reinforcement learning from human feedback. *arXiv preprint arXiv:2310.12773*.

Javier Ferrando, Oscar Obeso, Senthoran Rajamanoharan, and Neel Nanda. 2024. Do i know this entity? knowledge awareness and hallucinations in language models. *arXiv preprint arXiv:2411.14257*.

Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, and 1 others. 2022. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *arXiv preprint arXiv:2209.07858*.

Seungju Han, Kavel Rao, Allyson Ettinger, Liwei Jiang, Bill Yuchen Lin, Nathan Lambert, Yejin Choi, and Nouha Dziri. 2024a. Wildguard: Open one-stop moderation tools for safety risks, jailbreaks, and refusals